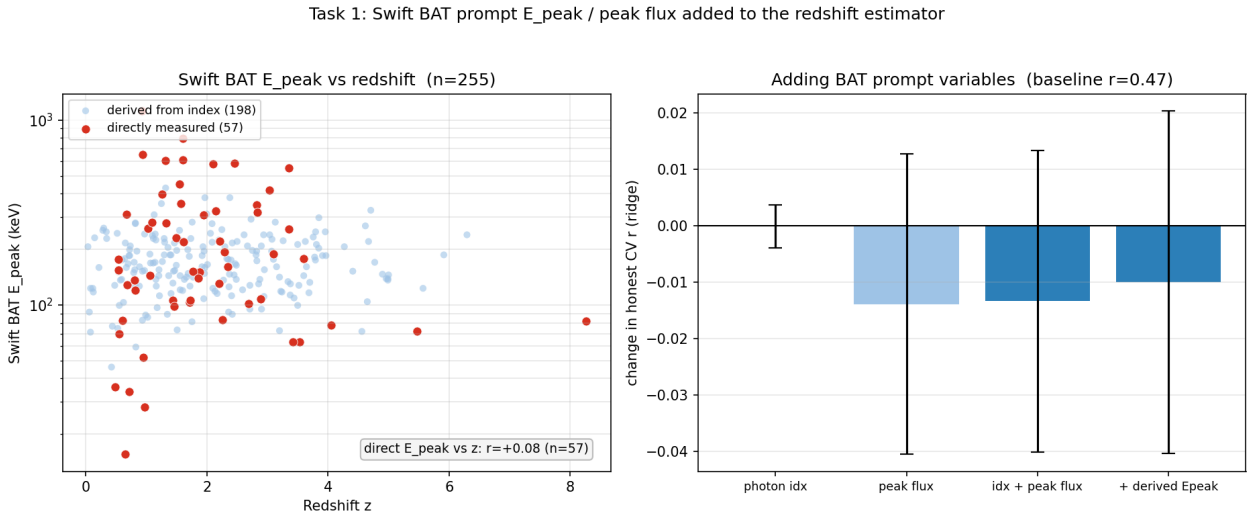
GRB Redshift Estimator: Next-Steps Results

Swift BAT E_peak and Fermi GBM substitution (figures) · October 2026

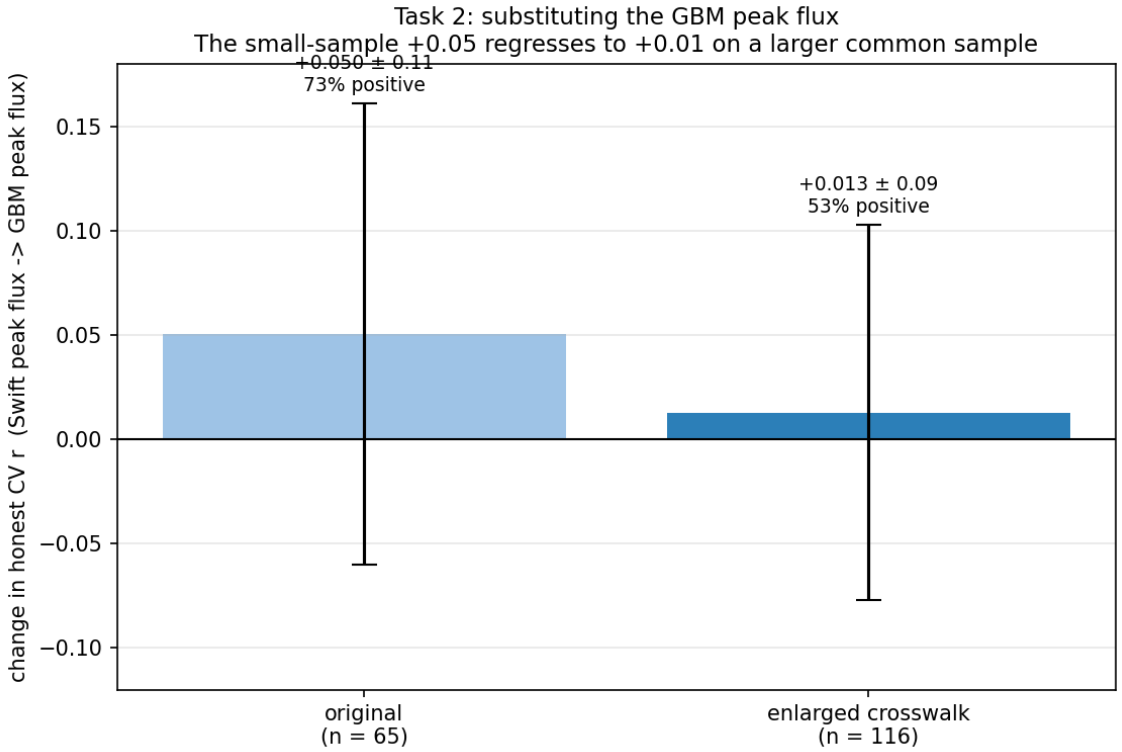
Summary. Two follow-up directions from Dr. Dainotti. **Task 1** gets a measured E_peak from the Swift BAT catalog and adds it to the estimator (no improvement -- a clean, non-circular confirmation that prompt E_peak carries little redshift signal). **Task 2** substitutes the Fermi GBM peak flux for the Swift one: a small-sample +0.05 did not survive a crosswalk-enlarged sample (it regressed to +0.01), so it is not a real gain. Neither task improves the estimator.

Task 1 — Adding a Swift-measured prompt E_peak to the redshift estimator



Left: the Swift BAT E_peak against redshift for the 255 sample GRBs with BAT data; the 57 directly-measured E_peak values (red) are shown separately from those derived from the photon index (light blue). The directly-measured E_peak is uncorrelated with redshift (r = +0.08, not significant). Right: the change in honest cross-validated r when BAT prompt variables are added to the estimator (baseline r = 0.47); error bars show the spread across cross-validation resamples. All changes are within the noise.

Task 2 — Using Fermi GBM prompt variables in place of Swift's



The change in honest cross-validated r when the Swift peak flux is replaced by the Fermi GBM peak flux. On the original 65-GRB common sample it was +0.05 (within the noise); on a sample nearly doubled by a date+position crosswalk to the Fermi GBM catalog (116 GRBs, with a healthier baseline r = 0.41), it regresses to +0.01 and is positive in only 53% of resamples -- the small-sample hint does not hold up.